

CASCADE

A Closed-Loop AI for Event-Driven Gridlock in Bengaluru

We do not just **forecast** the gridlock — we compute and **dispatch** the exact physical **resources** (officers, barricades, dynamic diversions) needed to **neutralize the congestion spike before it manifests**.

GCN+GRU trunk · DeepHit survival · Hawkes contagion · Conformal guarantees ·
OR-Tools & RL dispatch

Team CASCADE

Flipkart × BTP · Gridlock Hackathon 2.0 · Round 2

The Bengaluru Fallacy: Prediction Without Prescription

The Problem — Event-Driven, Cascading Gridlock

Bengaluru's paralysis is **event-driven**: one breakdown, a stadium let-out, a VIP movement or a water-logged underpass **cascades** into city-wide gridlock. The BTP must place **scarce physical resources**—roughly **10 tow-trucks** plus officers and barricades across **150+ hotspots**—*before* the spike materializes.

Why Pure Prediction Fails

- A heatmap is a **description, not a decision** — it says traffic is bad, never *what to do*.
- Dozens of teams forecast congestion; **none place a single officer**.
- They ignore **censoring** (12.3% of incidents still active) and **contagion**.

The Operational Gap

- City authorities don't need *another forecast*.
- They need an **actionable remedy** attached to a **confidence guarantee**.
- Passive analytics → **active operational control**.

The Paradigm Shift: A Closed-Loop Control System

“We don’t forecast the mess — **we neutralize it.**”



- **Predict:** the next incident’s location, timing & duration, each with **calibrated confidence**.
- **Prescribe:** the *exact* officers, barricades and diversions that relieve it.
- **Dispatch:** optimal coordinates + routes, **pre-positioned** ahead of the spike.
- **Learn:** de-censoring + off-policy reward **close the loop** on every outcome.

Intuition: treat the city as a **controllable system**, not a crystal ball.

Architecture: One Brain, Many Decisions

Shared Trunk — “City-State Encoder”

GCN (spatial: the 294-junction road graph, kNN + corridor edges) + **GRU** (temporal: recent incident history) → a state embedding H_t . A **Mamba** selective-SSM kernel is a validated drop-in; GRU remains the default for short ($L=8$) histories.

Head 1 — DeepHit Survival

Duration PMF over 20 bins + $P(\text{closure})$ + priority,
censoring-aware:

$$\mathcal{L}_{\text{nll}} = \begin{cases} -\log p_{k(i)} & \text{uncensored} \\ -\log \sum_{k > k(i)} p_k & \text{censored} \end{cases}$$

Head 2 — Hawkes Intensity

Self-exciting point process forecasts **the next incident**:

$$\lambda(t, s) = \mu_s + \sum_{t_j < t} \alpha e^{-\beta(t-t_j)}$$

μ_s baseline + **excitation** that decays after each nearby event = **contagion**.

Fusion & Event-Driven Volatility

Heads share the trunk via **MMoE** with **GradNorm**→**PCGrad** gradient surgery (kills negative transfer) and Kendall uncertainty weights. High-volatility events—**VIP movement, processions, matches, public events, water-logging**—enter as causal/categorical features; a conditional **DDPM** **augments the rare 6% planned-event class** so the model never starves, and the Hawkes excitation captures the **cascade** from a stadium let-out.

The Dispatch Engine: Math That Places Officers

Trust Layer — Conformal + ACI

Every clearance estimate gets a **coverage-guaranteed** P90 bound; **Adaptive Conformal Inference** holds it under drift:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha_{\text{target}} - \text{err}_t)$$

90.7% empirical coverage on held-out data.

Allocator — OR-Tools CP-SAT

Maximal-Coverage for officers + budgeted **ILP** for barricades:

$$\max \sum_i d_i x_i \quad \text{s.t.} \quad x_i \leq \sum_{j \in N_R(i)} y_j, \quad \sum_j y_j \leq P$$

d_i = severity-minutes; N_R within response radius.

From Prediction to Deterministic Resource Allocation

- **Predictive pre-positioning:** demand = $(1 - w) \cdot \text{current burden} + w \cdot \text{Hawkes intensity} \times \text{horizon}$ — staff *before* the spike.
- **GATE-PPO** graph-encoder RL dispatcher beats the greedy baseline by +20.3% congestion relieved.
- **Equity constraint** breaks the predictive-enforcement feedback loop; barricade ILP scored by $P(\text{closure}) \times \text{spillover}$.
- Output is **physical and actionable**: officer lat/long + head-counts, ranked barricades, networkx diversion routes.

The Command Center: Built for the Control Room

The Hero — De-Biased 3D City Map

GPU-rendered **3D extruded congestion hexbins** over labelled Bengaluru (deck.gl on CARTO/Mappls). A one-click **naive** ↔ **corrected** toggle separates *where patrols logged tickets* from *where impact concentrates* — the project's core insight, shown live.

Interactive Operational Control

- **What-if sliders** — officer teams, barricades, response radius, predictive weight — **re-run OR-Tools live**.
- **Time-of-day simulation** sweeps demand across the network.
- Focus-junction inspector with calibrated P50/P90 bounds.

Decision Intelligence

- **Commander's Briefing (XAI)** translates the optimization into plain English.
- DeepHit survival curve + Hawkes risk gauges.
- **Live Network Pulse** — active incidents, units engaged, avg clearance, incident ticker.

Four tabs: **Command Center** · Model Transparency · Economic Impact · Deployment Worklist.

Viability, Scale & Measured Impact

Validated Performance (held-out test)	
Metric	Result
Survival C-index (verified labels)	0.667 (GCN) vs. 0.663 XGBoost; ensemble 0.670
Calibration (Integrated Brier)	0.0514 vs. 0.0546 (XGBoost AFT)
Conformal coverage (target 90%)	90.7% empirical · 90.0% under drift (ACI)
Hawkes next-incident C-index	0.645 (no baseline can do this)
RL dispatcher vs. greedy	+20.3% congestion relieved
Plan vs. random (realized)	+360%
Digital-twin impact (illustrative)	~51% congestion avoided · ~Rs. 4.9M/day

Deployment & Efficiency

- Single **Streamlit** app, **Dockerized**, one-URL on HF Spaces / Render.
- Reads **precomputed artifacts** → sub-second; live re-optimization in an isolated OR-Tools subprocess.
- Deep training on **one Colab GPU**; RL, ETL & optimization run on **CPU**.

Real-World Fit

- Maps to BTP’s **real constraint** (≈10 tow-trucks, 294 junctions).
- **Honest reporting** — both full & verified-label slices.
- Built **only on the provided ASTRAM log**; Mappls used as an allowed *service*.

CASCADE turns a passive incident log into a calibrated, contagion-aware dispatch system — a trustworthy plan for the next hour.